

* Pure Spinor in GS: $S_c = \int d\tau (p^\mu \dot{x}_\mu + \omega_\alpha \nabla \lambda^\alpha - \frac{1}{2} L^\alpha (\not{P} \lambda)_\alpha)$
 $\nabla \lambda^\alpha = \dot{\lambda}^\alpha + A \lambda^\alpha$

$$\lambda \gamma^\mu \lambda = 0, \quad \omega_\alpha \simeq \omega_\alpha + \Lambda^\mu (\gamma_\mu \lambda)_\alpha$$

Scale symmetry: $\lambda^\alpha \rightarrow \Omega \lambda^\alpha, \quad \omega_\alpha \rightarrow \Omega^{-1} \omega_\alpha$

$$L^\alpha \rightarrow \Omega^{-1} L^\alpha, \quad A \rightarrow A - \frac{\dot{\Omega}}{\Omega}$$

$$\theta^\alpha \rightarrow \Omega^{-1} \theta^\alpha, \quad \rho \rightarrow \Omega^{-1} \rho, \quad \varphi \rightarrow \Omega^{-2} \varphi$$

BRST: $\left\{ \begin{array}{l} Q L^\alpha = \nabla \theta^\alpha - \rho \lambda^\alpha, \quad Q \rho = \nabla \rho \\ Q x^\mu = \frac{1}{2} \lambda \gamma^\mu \theta, \quad Q \omega_\alpha = -\frac{1}{2} (\not{P} \theta)_\alpha \cdot \varphi \frac{\delta \chi}{\delta L^\alpha} \\ Q \theta^\alpha = \varphi \lambda^\alpha, \quad Q \beta = N, \quad Q N = 0 \end{array} \right\}$

$(\not{P} \lambda)_\alpha$: Twistor-line constant, which replaces the on-shell condition

* $S = S_c + \int d\tau Q \chi$

$$\chi = \rho_\alpha L^\alpha + \beta \rho \quad (\text{Gauge fixing fermion})$$

→ In the Gauge $\varphi = 1$

$$S = \int d\tau (p_\mu \dot{x}^\mu + \omega_\alpha \dot{\lambda}^\alpha - \rho_\alpha \dot{\theta}^\alpha)$$

* To obtain GS we fix χ_{GS} s.t. $Q \omega_\alpha = 0$

$$\delta \chi / \delta L^\alpha = -\frac{1}{2} (\not{P} \theta)_\alpha$$

$$\chi_{GS} = A - L^\alpha (\frac{1}{2}) (\not{P} \theta)_\alpha \varphi^{-1}$$

to retain BRST exact forms $A = \beta \rho$

$$\chi_{GS} = -\varphi^{-1} \frac{1}{2} L^\alpha (\not{P} \theta)_\alpha + \beta \rho$$

$$Q \chi_{GS} = -\rho \frac{1}{2} (\nabla \theta^\alpha - \rho \lambda^\alpha) (\not{P} \theta)_\alpha + \varphi^{-1} (\frac{1}{2}) L^\alpha (\not{P} \varphi \lambda_\alpha) + \beta (\nabla \rho) + N \rho$$

$$\Rightarrow S = S_{GS} + \int d\tau Q \chi_{GS}$$

$$= \int d\tau \left[p_\mu \dot{x}^\mu + \omega_\alpha \nabla \lambda^\alpha - \frac{1}{2} L^\alpha (\not{P} \lambda)_\alpha - \frac{1}{2} \varphi^{-1} (\nabla \theta^\alpha) (\not{P} \theta)_\alpha \right.$$

$$\left. + \frac{1}{2} \lambda^\alpha (\not{P} \theta)_\alpha + \frac{1}{2} \varphi^{-1} L^\alpha (\not{P} \lambda)_\alpha + \beta (\nabla \rho) + N \rho \right]$$

$$S = \int d\tau \left[p_\mu \dot{x}^\mu + \omega_\alpha \nabla \lambda^\alpha - L^\alpha (\not{P} \lambda)_\alpha + \beta \nabla \rho + N \rho + \hat{\mathcal{E}} \right]$$

• Gauge fix $\varphi = 1$

E.O.M $\omega_\alpha A^\alpha + 2\beta A^\varphi \rightarrow \omega_\alpha \lambda^\alpha A + \beta A$

$B: A = 0 \quad \ominus \quad \varphi = 1$

$A: \beta = -\frac{1}{2} \omega_\alpha \lambda^\alpha$

$S = \int d\tau \left(P_m \dot{x}^m + \omega_\alpha \lambda^\alpha - L^\alpha (\not{P}\lambda)_\alpha - \frac{1}{2} \dot{\Theta}^\alpha (\not{P}\Theta)_\alpha \right)$

$\omega_\alpha \lambda^\alpha = Q(\omega_\alpha \dot{\Theta}^\alpha)$

→ Add to χ_{GS} to cancel term

$\chi_{GS} = -\frac{1}{2} L^\alpha (\not{P}\Theta)_\alpha \varphi^{-1} + \beta \varphi - \omega_\alpha (\not{P}\Theta)^\alpha \varphi^{-1}$

→ $S_{GS} = \int d\tau P_m \underbrace{\left(\dot{x}^m - \frac{1}{2} \dot{\Theta}^\mu \gamma^\mu \Theta \right)}_{\pi^m} - L^\alpha (\not{P}\lambda)_\alpha$

$S_{GS} = \int d\tau \left(P_m \pi^m - L^\alpha (\not{P}\lambda)_\alpha \right)$

* Would like to see $\delta\pi = Q\pi = 0$

$\begin{cases} Q\Theta^\alpha = \lambda^\alpha, & QL^\alpha = \dot{\Theta}^\alpha, & Qx^m = \frac{1}{2} \lambda \gamma^m \Theta \\ QP^m = 0 \end{cases}$

E.O.M: $(\not{P}\lambda)_\alpha = 0 \Rightarrow P^\mu P_\mu = 0$

• Can then use $\not{P}$ as a projection operator and have, $P_1 + P_2 = \mathbb{1}$ with $\not{P}P_1 = 0$ and $\not{P}P_2 = \not{P}$

⇒ A solution to the above is to have

$\lambda_\alpha = (\not{P}K)_\alpha \in V_1$

Claim: The Noether charge of the above symmetry is zero $Q^{Noether} = 0$

proof. $\frac{\partial Q}{\partial \dot{x}^m} Qx^m + \frac{\partial Q}{\partial \dot{\Theta}^\alpha} Q\Theta^\alpha = 0$

$= P_m \left(\frac{1}{2} \lambda \gamma^m \Theta \right) + \left(P_m \left(-\frac{1}{2} \right) \gamma_{\mu\beta} \Theta^\beta \right) \lambda^\mu$

$= 0$ so long as $[\Theta, \lambda] = 0$

→ $\mathcal{J} = 0$ implies the symmetry is actually local and we have not fully gauge fixed the symmetry.

• Note that this is evident from the "free" D.O.F. in K be arbitrary i.e. 2 fermionic D.O.F. which have not be accounted for.

• In fact, this is simply K -symmetry in the GS formalism.

$$\delta L^\alpha = \delta \theta^\alpha \rightarrow \delta(-L^\alpha \mathcal{P} \lambda_\alpha) = -\delta \mathcal{P} \lambda = (-\delta \theta^\alpha \kappa_\alpha) \mathcal{P}^2$$

$L^\alpha \rightarrow e$ replaces the vielbein

* Methods work for strings and membranes.

Twistor like string (ΠB - strings) $D=10$

• first order form: $\alpha, \hat{\alpha} = 1, \dots, 16$ $SO(8)$

$$\begin{array}{ccc} x^m & \lambda^\alpha & \hat{\lambda}^{\hat{\alpha}} \\ p_m & \omega_\alpha & \omega^{\hat{\alpha}} \end{array} \quad \begin{array}{l} \text{opposite chirality} \\ \lambda \gamma^m \lambda = 0 \text{ and } \hat{\lambda} \gamma^m \hat{\lambda} = 0 \end{array}$$

• Scale symmetry also independent on both sides.

(projective 2)
spinors

$$\begin{cases} \lambda^\alpha \rightarrow \Omega \lambda^\alpha \\ \hat{\lambda}^{\hat{\alpha}} \rightarrow \tilde{\Omega} \hat{\lambda}^{\hat{\alpha}} \end{cases}$$

Constraints:

$$\begin{aligned} (p^m + \partial_\sigma x^m) (\gamma_m \lambda)_\alpha &= 0 \\ (p^m - \partial_\sigma x^m) (\gamma_m \hat{\lambda})_{\hat{\alpha}} &= 0 \end{aligned}$$

$$\begin{aligned} \hookrightarrow (p + \partial_\sigma x)^2 &= 0 & L & \text{-moving Virasoro constraints} \\ (p - \partial_\sigma x)^2 &= 0 & R & \end{aligned}$$

$$T_{00} = 0$$

$$I \subset (\sigma, \tau)$$

we need to add

$$\begin{aligned} \nabla_\pm \lambda^\alpha &= 0 \\ \nabla_\pm \hat{\lambda}^{\hat{\alpha}} &= 0 \end{aligned}$$

$$\nabla_\pm \lambda^\alpha \equiv \partial_\pm \lambda^\alpha + A_\pm \lambda^\alpha$$

$$\nabla_\pm \hat{\lambda}^{\hat{\alpha}} \equiv \partial_\pm \hat{\lambda}^{\hat{\alpha}} + \tilde{A}_\pm \hat{\lambda}^{\hat{\alpha}}$$

$$S_C = \int d\tau d\sigma \left[p_m \dot{x}^m + \omega_\alpha \nabla_\pm \lambda^\alpha + \tilde{\omega}_{\hat{\alpha}} \nabla_\pm \hat{\lambda}^{\hat{\alpha}} + L(p - \partial_\sigma x) \lambda + \tilde{L}(p - \partial_\sigma x) \hat{\lambda} + K_\alpha \nabla_\pm \lambda^\alpha + \tilde{K}_{\hat{\alpha}} \nabla_\pm \hat{\lambda}^{\hat{\alpha}} \right]$$

$\Rightarrow \lambda^\alpha$: If λ^α has a constant direction then the system mimics ~ topological string: $\lambda \in \hat{\lambda}$ can choose two complex structures.

• Here λ has no σ dependence.

$\Rightarrow$ To make the d, p, f, eo symm $\rightarrow$ Second order form

$\sim c \left(\frac{1}{2} (p + \partial_\alpha x)^2 - 2\omega_\alpha \nabla_\alpha \lambda^\alpha \right) + \bar{c} \left(\frac{1}{2} (p - \partial_\alpha x)^2 - 2\bar{\omega}_\alpha \nabla_\alpha \bar{\lambda}^\alpha \right)$

$\underbrace{\hspace{10em}}_{cT} \quad \underbrace{\hspace{10em}}_{\bar{c}\bar{T}}$

which leaves the action invariant: can simply shifting

$$\begin{aligned}
 L^\alpha &\rightarrow L^\alpha - (\phi + \partial_\alpha x) \frac{\bar{\lambda}}{\lambda \bar{\lambda}} e \\
 \hat{L}^{\hat{\alpha}} &\rightarrow \hat{L}^{\hat{\alpha}} - (\phi - \partial_\alpha x) \frac{\bar{\lambda}}{\lambda \bar{\lambda}} e
 \end{aligned}$$

(B-ghost shift)

$$\begin{aligned}
 K_\alpha &\rightarrow K_\alpha + 2\omega_\alpha \\
 \hat{K}_{\hat{\alpha}} &\rightarrow \hat{K}_{\hat{\alpha}} + 2\bar{\omega}_{\hat{\alpha}}
 \end{aligned}$$

$$\begin{aligned}
 S_c = \int d\tau d\sigma \left[e \left[-\frac{1}{2} \nabla x^m \tilde{\nabla} x_m + \omega_\alpha \nabla \lambda^\alpha + \bar{\omega}_{\hat{\alpha}} \nabla \hat{\lambda}^{\hat{\alpha}} + \frac{1}{2} \nabla x^m (L \gamma_m \lambda) \right. \right. \\
 \left. \left. + K_\alpha \nabla \lambda^\alpha + \frac{1}{2} \nabla x^m (\hat{L} \gamma_m \hat{\lambda}) + \hat{K}_{\hat{\alpha}} \nabla \hat{\lambda}^{\hat{\alpha}} - \frac{1}{4} (L \gamma^m \lambda) (\hat{L} \gamma_m \hat{\lambda}) \right] \right]
 \end{aligned}$$

$$e = \begin{pmatrix} 1 & +2e \\ -2\bar{e} & 1 \end{pmatrix}, \quad U(1) \text{ invariant } |e| = (I - 4\bar{e}e)$$

• Coulomb gauge: $e = \bar{e} = 0$
 • lightcone gauge: $x^+ = \tau$

$$\begin{aligned}
 S = \int d\tau d\sigma \left(p_m \dot{x}^m + \omega_\alpha \nabla_\tau \lambda^\alpha + \bar{\omega}_{\hat{\alpha}} \nabla_\tau \hat{\lambda}^{\hat{\alpha}} + L(\phi + \partial_\sigma x) \lambda \right. \\
 \left. + K_\alpha \nabla_\tau \lambda^\alpha + \hat{K}_{\hat{\alpha}} \nabla_\tau \hat{\lambda}^{\hat{\alpha}} + \hat{L}(\phi - \partial_\sigma x) \hat{\lambda} \right)
 \end{aligned}$$

$$\begin{aligned}
 K: K_\alpha \nabla_\tau \lambda^\alpha &\rightarrow (K_\alpha + 2\omega_\alpha) \nabla_\tau \lambda^\alpha \\
 \hat{K}_{\hat{\alpha}} \nabla_\tau \hat{\lambda}^{\hat{\alpha}} &\rightarrow (\hat{K}_{\hat{\alpha}} + 2\bar{\omega}_{\hat{\alpha}}) \nabla_\tau \hat{\lambda}^{\hat{\alpha}}
 \end{aligned}$$

$$L: L^\alpha = \left((\phi - \partial_\sigma x) \frac{\bar{\lambda}}{\lambda \bar{\lambda}} \right)^\alpha e$$

$$\begin{aligned}
 L(\phi - \partial_\sigma x) \hat{L} &\rightarrow L(\phi - \partial_\sigma x) \lambda - \left((\phi + \partial_\sigma x) \frac{\bar{\lambda}}{\lambda \bar{\lambda}} \right) (\phi + \partial_\sigma x) \lambda e \\
 &\bullet (p_m + \partial_\sigma x_m) \gamma^m{}_{\alpha\beta} \bar{\lambda}_\beta (x_n)_{\alpha\delta} (p^n + \partial_\sigma x^n) \lambda^\delta \frac{1}{\lambda \bar{\lambda}} e \\
 &= (p_m + \partial_\sigma x_m) (2\delta_n^m \delta_\sigma^B) \bar{\lambda}_B (p^n + \partial_\sigma x^n) \lambda^\delta e \\
 &= 2(p_m p^m + 2p^m \partial_\sigma x_m + \partial_\sigma x^m \partial_\sigma x_m) \bar{\lambda}_B \lambda^\delta / \langle \lambda \bar{\lambda} \rangle e \\
 &= 2(p_m p^m + 2p^m \partial_\sigma x_m + \partial_\sigma x^m \partial_\sigma x_m)
 \end{aligned}$$

$$\sim 2(p^2 + 2p^\mu \partial_\mu x_m + \partial_\mu x^m \partial_\mu x_m) e$$

$$\hat{L} : \sim 2(p^2 - 2p^\mu \partial_\mu x_m + \partial_\mu x^m \partial_\mu x_m) \bar{e}$$

Adding these terms

$$S_c = \int d\tau d\sigma \left\{ p^\mu \dot{x}_m + \omega_\alpha \nabla_I \lambda^\alpha + \tilde{\omega}_{\dot{\alpha}} \bar{\nabla}_I \hat{\lambda}^{\dot{\alpha}} - L(\not{p} + \not{\partial} x) \lambda \right. \\ \left. + \kappa_\alpha \nabla_I \lambda^\alpha - \hat{L}(\not{p} - \not{\partial} x) \hat{\lambda} + \tilde{\kappa}_{\dot{\alpha}} \bar{\nabla}_I \hat{\lambda}^{\dot{\alpha}} \right. \\ \left. + e \left(\frac{1}{2} (\not{p} + \not{\partial} x)^2 + 2\omega_\alpha \nabla_I \lambda^\alpha \right) \right. \\ \left. + \bar{e} \left(\frac{1}{2} (\not{p} - \not{\partial} x)^2 - 2\tilde{\omega}_{\dot{\alpha}} \bar{\nabla}_I \hat{\lambda}^{\dot{\alpha}} \right) \right\}$$

P.E.O.M.:

$$\dot{x}^m + \frac{1}{2} e (p^m + 2\partial_\sigma x^m) + L^\alpha \gamma_{\alpha\beta}^m \lambda^\beta + \hat{L}^{\dot{\alpha}\dot{\beta}m} \hat{\lambda}^{\dot{\alpha}} \\ + \frac{1}{2} \bar{e} (p^m - 2\partial_\sigma x^m) = 0$$

$$\frac{1}{2} p^m (e + \bar{e}) + \dot{x}^m + \partial_\sigma x^m (e - \bar{e}) - L \gamma \lambda - \hat{L} \hat{\lambda} = 0$$

$$p^m = 2(-1)(e + \bar{e})^{-1} [\dot{x}^m + \partial_\sigma x^m (e - \bar{e}) + L + \hat{L}]$$

For stress $L \equiv L \gamma \lambda$ and $\hat{L} \equiv \hat{L} \gamma \hat{\lambda}$

$$\Rightarrow (i) 2[(-1)(e + \bar{e})^{-1}] [\dot{x}^m + \partial_\sigma x^m (e - \bar{e}) + L + \hat{L}] \dot{x}_m$$

$$= 2[(-1)(e + \bar{e})^{-1}] [\dot{x}^m \dot{x}_m + \partial_\sigma x^m \dot{x}_m (e - \bar{e}) + L \dot{x}_m + \hat{L} \dot{x}_m]$$

$$(ii) L^\alpha \gamma_{\alpha\beta}^m (2(-1)(e + \bar{e})^{-1}) [\dot{x}^m + \partial_\sigma x^m (e - \bar{e}) + L + \hat{L}]$$

$$+ \partial_\sigma x_m \} \lambda^\beta$$

$$= \beta L \gamma \lambda \dot{x}_m + \beta L \gamma \lambda \partial_\sigma x_m (e - \bar{e}) + L^2 \dots$$

$$= \beta L \gamma \lambda (\dot{x}_m + \beta (e - \bar{e}) \partial_\sigma x_m)$$

$$+ L^\alpha \gamma_{\alpha\beta}^m \lambda^\beta \beta (L^\gamma \gamma_{\gamma\delta}^m \lambda^\delta) + \beta L^\alpha \gamma_{\alpha\beta}^m \lambda^\beta (\hat{L}^{\dot{\alpha}\dot{\beta}m} \hat{\lambda}^{\dot{\alpha}})$$

Adding Momentum terms:

$$\frac{1}{2} (p^2 + 2p^\mu \partial_\mu x_m + \partial_\mu x^m \partial_\mu x_m) e$$

$$* p^2 = \beta^2 \left[\dot{x}^m \dot{x}_m + \dot{x}^m \partial_\sigma x_m (e - \bar{e}) + L \dot{x}_m + \hat{L} \dot{x}_m \right. \\ \left. + \dot{x}^m \partial_\sigma x_m + \partial_\sigma x^m \partial_\sigma x_m (e - \bar{e})^2 + L \partial_\sigma x_m + \hat{L} \partial_\sigma x_m \right. \\ \left. + \dot{x}_m L + \partial_\sigma x_m (e - \bar{e}) + L^2 + \hat{L} L \right. \\ \left. + \dot{x}_m \hat{L} + \partial_\sigma x_m (e - \bar{e}) \hat{L} + L \hat{L} + \hat{L}^2 \right]$$

$$\Rightarrow L \hat{L} = (L^\alpha \gamma_{\alpha\beta}^m \lambda^\beta) (\hat{L}^{\dot{\alpha}\dot{\beta}m} \hat{\lambda}^{\dot{\alpha}})$$

$$* 2p^\mu \partial_\mu x_m = 2\beta [\partial_\sigma x^m \dot{x}_m + \partial_\sigma x^m \partial_\sigma x_m (e - \bar{e}) + \partial_\sigma x^m L + \partial_\sigma x^m \hat{L}]$$

$$\Rightarrow \nabla_k \equiv e_k^\mu \nabla_\mu, \quad \bar{\nabla}_{\dot{k}} \equiv e_{\dot{k}}^{\dot{\mu}} \bar{\nabla}_{\dot{\mu}}$$

$\Rightarrow$

* After Gauge fixing $S = S_c + \int Q \chi$

$$- \int d^2z \left(-\frac{1}{2} \partial x^\mu \partial x_\mu + \omega_\alpha \partial \lambda^\alpha - p_\alpha \partial \hat{\theta}^\alpha \right)$$

Right moving: $T = -\frac{1}{2} \partial x^\mu \partial x_\mu + \omega_\alpha \partial \lambda^\alpha - p_\alpha \partial \hat{\theta}^\alpha$
 Since this is a free field theory we can calculate OPE $T(z)T(w)$

$\rightarrow \lambda$ is constrained by $\lambda \gamma^\mu \lambda = 0$

{	10 vectors $\partial x^\mu \partial x_\mu$	10	$\partial x^\mu \partial x_\mu$
	16 $\hat{\theta}^\alpha \times 2$ (fermionic)	32	
	22 Bosonic from ω_α	22	

$\Rightarrow C=0$ (has no conformal charge)

* The Lorentz spin generator of the theory:
 $M^{\mu\nu} = \psi^\mu \psi^\nu$ (for general Majorana in 10D)

$\rightarrow$ For pure spinors ($\lambda \gamma^\mu \lambda = 0$)

$$\tilde{M}^{\mu\nu} = \frac{1}{2} p^\alpha \gamma_{\alpha\beta}^{\mu\nu} \hat{\theta}^\beta + \frac{1}{2} \omega^\alpha \gamma_{\alpha\beta}^{\mu\nu} \lambda^\beta$$

Lorentz is fixed by the fermionic content of the theory.

OPE of $M_x^{\mu\nu} M_z^{\rho\sigma} \sim \underbrace{\frac{2\eta^{\mu\nu} \eta^{\rho\sigma}}{(y-z)^2}}_{\text{level of the theory}} + \underbrace{\frac{2\eta^{\rho\sigma} M^{\mu\nu}}{(y-z)}}_{\text{SO(9,1) Algebra}}$

• RNS String: $k=1$

• Pure Spinor: $k=1 \geq 4 + (-3)$

$\hookrightarrow$ Reduce $SO(1,9) \rightarrow SU(5)$

$\hookrightarrow$ Calculate OPE in the non-compact $SU(5)$ reps

$$\begin{aligned} \frac{1}{2} \omega^\alpha \gamma_{\alpha\beta}^{\mu\nu} \lambda^\beta &\equiv N^{\mu\nu} \\ \frac{1}{2} \omega^\alpha \gamma_{\alpha\beta}^{\rho\sigma} \lambda^\beta &\equiv N^{\rho\sigma} \end{aligned} \quad \left| \quad \begin{aligned} J &\equiv \omega_\alpha \lambda^\alpha \\ &= -3\eta^{\mu\nu} \eta^{\rho\sigma} \end{aligned} \right. \quad \begin{pmatrix} \lambda^\alpha \\ \lambda^{\alpha\beta} \\ \lambda \end{pmatrix}$$

$$\Rightarrow \text{OPE } N_{(y)}^{\mu\nu} N_{(z)}^{\rho\sigma} \sim \frac{-3\eta^{\mu\nu} \eta^{\rho\sigma}}{(y-z)^2} + \frac{\eta^{\rho\sigma} N^{\mu\nu}}{(x-z)}$$

$\underline{u} = -3$

* Right movers is equivalent $\bar{E}\bar{T}$

$\rightarrow$ To get physical states: need BRST operator:

$$Q = \oint dz \lambda^\alpha d_\alpha$$

$$* d\alpha = p_\alpha + \frac{1}{2}(\partial x^m (\gamma_m \theta)_\alpha - \frac{1}{4}(\gamma_m \theta)_\alpha (\partial_\theta \gamma^m \theta))$$

* (Paper has $\frac{1}{2} p^m (\gamma_m \theta)_\alpha$ term but I've already gauged this away) *

OPE of $d\alpha$ are trivial:

$$d_\alpha(y) d_\beta(z) \sim \frac{1}{(y-z)} \gamma^m_{\alpha\beta} \pi_m$$

$$\pi^m = (\partial x^m - \frac{1}{2} \partial_\theta \gamma^m \theta)$$

$$x^m(y) x^n(z) \sim -\log|z-y| \delta^{mn}$$

$$p_\alpha(y) \Theta^\beta(z) \sim \delta_\alpha^\beta \frac{1}{y-z}$$

W_α : Gauge invariant. $W_\alpha \sim W_\alpha + \lambda^m (\gamma_m \lambda)_\alpha$

Open Vertex operators: V , $QV=0$

$$V \sim V + Q\Omega$$

Has the same vertex operator as the pure Spiver superparticle.

$$* V = \lambda^\alpha A_\alpha(x, \theta) *$$

• For Massless states this is simply SYM

$\int d^2z \mathcal{U}$ is BRST invariant if $Q\mathcal{U} = \partial V$

→ For massless case (SYM)

$$\mathcal{U} = \partial \theta^\alpha A_\alpha + \pi^m A_m + d_\alpha W^\alpha + \frac{1}{2} N_{mn} F^{mn}$$

$$A_m = \gamma_m^{\alpha\beta} D_\alpha A_\beta$$

$$W^\alpha = D^\alpha A = \partial_m A_\alpha \gamma^{m\alpha}$$

$$F = DA$$

$$\rightarrow A_m \sim \gamma_m^{\alpha\beta} \left(\frac{1}{\partial \theta^\alpha} + \frac{1}{2} (\gamma^m \theta)_\alpha \partial_m \right) A_\beta$$

$$\sim \gamma_m^{\alpha\beta} \left(\frac{1}{2} \frac{1}{\partial \theta^\alpha} (\gamma^m)_{\beta\gamma} \theta^\gamma \partial_m \right) A_\alpha$$

$$\sim \gamma_m^{\alpha\beta} \frac{1}{2} (\gamma^m)_{\beta\alpha} A_\alpha$$

$$\sim 2 \left(\frac{1}{2} \right) \text{Tr}(\gamma_m \gamma^n) A_n$$

$$\sim \dim(\text{Spin}(1, D-1)) A_m + \dots$$

* Claim: $QV = \partial V$

$$QV = \partial(\lambda^\alpha A_\alpha) + \lambda^\alpha \partial_\theta^\beta (-D_\alpha A_\beta - D_\beta A_\alpha + \gamma_{\alpha\beta}^m A_m) + \lambda^\alpha \pi^m (D_\alpha A_m - \partial_m A_\alpha - \gamma_{m\alpha\beta} W^\beta) + \dots$$

* Note: Paper has B_m but this is simply

$$B_m = \gamma_m^{\alpha\beta} D_\alpha A_\beta$$

Proof:

$$Q = \int dz \lambda^\alpha \left[p_\alpha + \frac{1}{2} (\partial x^m) (\gamma_m \theta)_\alpha - \frac{1}{4} (\gamma^m \theta)_\alpha (\partial \theta \gamma_m \theta) \right]$$

$$= \int dz \lambda^\alpha \left[p_\alpha + \frac{1}{2} (\gamma_m \theta)_\alpha (\partial x^m - \frac{1}{2} \partial \theta \gamma^m \theta) \right]$$

$$= \int dz \lambda^\alpha \left[p_\alpha + \frac{1}{2} (\gamma_m \theta)_\alpha \pi^m \right]$$

(i) $\frac{1}{2} \lambda^\alpha (\gamma_m \theta)_\alpha \pi^m \partial^\alpha A_\alpha$

(ii) $\pi_m \pi^m = (\partial x^m - \frac{1}{2} \partial \theta \gamma^m \theta) (\partial x_m - \frac{1}{2} \partial \theta \gamma_m \theta)$

$$= \partial x^m \partial x_m - \partial x^m \partial \theta \gamma_m \theta + \frac{1}{4} (\partial \theta \gamma_m \theta)^2$$

$$(\partial \theta^\alpha \gamma_{\alpha\beta} \theta^\beta) (\partial \theta^\gamma \gamma_{\gamma\delta} \theta^\delta) = \partial \theta^\alpha (2 \delta_{\alpha\gamma} \epsilon_{\beta\delta}) \theta^\beta \partial^\gamma \theta^\delta$$

$$= 2 \partial \theta^\alpha \partial_\alpha \theta^\beta \theta_\beta$$

Third term:

$$\lambda^\alpha \left(p_\alpha + \frac{1}{2} \partial x^m (\gamma_m \theta)_\alpha + \frac{1}{4} (\gamma^m \theta)_\alpha (\partial \theta \gamma_m \theta) \right) \left[-D_\alpha W^\beta + \frac{1}{4} (\gamma^{mn})_\alpha{}^\beta F_{mn} \right]$$

$$\left(D_\alpha W^\beta = D_\alpha (D_\gamma A_m - D_m A_\gamma) (\gamma^m)^\beta{}_\gamma \right)$$

$$\lambda^\alpha \left(p_\alpha + \frac{1}{2} \partial x^m (\gamma_m \theta)_\alpha + \frac{1}{4} (\gamma^m \theta)_\alpha (\partial \theta \gamma_m \theta) \right) \left[-D_\alpha (D_\gamma A_m - D_m A_\gamma) (\gamma^m)^\beta{}_\gamma + \frac{1}{4} (\gamma^{mn})_\alpha{}^\beta F_{mn} \right]$$

$$\rightarrow F_{mn} = \partial_{[m} A_{n]} = \gamma_{[m}{}^\alpha{}_\beta \partial_{n]} D_\alpha A_\beta$$

$$= \lambda^\alpha \left[p_\alpha + \frac{1}{2} \partial x^m (\gamma_m \theta)_\alpha + \frac{1}{4} (\gamma^m \theta)_\alpha (\partial \theta \gamma_m \theta) + \frac{1}{4} (\gamma^{mn})_\alpha{}^\beta F_{mn} \right]$$

* For the closed string: LR product of the open Tree amplitude Scattering:

$$a = \langle V_1 V_2 V_3 \rangle \langle z_1 z_2 \dots z_n \rangle$$

$$V = \sigma_m(x) \partial x^m + \frac{1}{2} \partial \gamma_m a_n M^{mn} + \zeta^\alpha(x) q_\alpha + \mathcal{O}(\theta^2)$$

$$M^{mn} = \frac{1}{2} p \gamma^{mn} \theta + N^{mn}$$

$$q_\alpha = p_\alpha + \frac{1}{2} (\partial x^m + \frac{1}{2} \theta \gamma^m \partial \theta) (\gamma_m \theta)_\alpha$$

(Spacetime SUSY gen)

For Type II: $\mathcal{U} = \lambda^\alpha \hat{\lambda}^\beta A_{\alpha\beta}(x, \theta, \hat{\theta})$

$$Q\mathcal{U} = \hat{Q}\mathcal{U} = 0$$

$$\hookrightarrow Q = \int \lambda^\alpha d\alpha \quad \hat{Q} = \int \hat{\lambda}^\alpha d\hat{\alpha}$$

Gauge Condition: $\partial \mathcal{U} = Q \cdot \hat{\Omega} + \hat{Q} \Omega, \quad \hat{Q} \Omega = Q \cdot \Omega = 0$

$$1 \leq \hat{\alpha} \leq n = \int \lambda^\beta d\beta \int \lambda^\alpha \hat{\lambda}^\beta A_{\alpha\beta}(\hat{\lambda}) = 0$$

$$= \int \lambda^B (p_B + \frac{1}{2} (\gamma_m \theta)_B \pi^m) \lambda^\alpha \hat{\lambda}^{\tilde{B}} A_{\alpha \tilde{B}}$$

~> Look at @ term:

$$\begin{aligned} & (-\frac{1}{4}) \lambda^B (\gamma_m)_B \gamma^\theta \partial_\theta (\gamma^m)_{\delta\rho} \theta^\rho \lambda^\alpha \hat{\lambda}^{\tilde{B}} A_{\alpha \tilde{B}} \\ & \sim (-\frac{1}{4}) \lambda^B (\gamma_m)_B \gamma^\theta \partial_\theta (\gamma^m)_{\delta\rho} \theta^\rho \epsilon^{\alpha\tilde{\nu}} \lambda_{\tilde{\nu}} \hat{\lambda}^{\tilde{B}} A_{\alpha \tilde{B}} \\ & \sim (-\frac{1}{4}) \lambda^B (\gamma_m)_B \gamma^\theta \partial_\theta (\gamma^m)_{\delta\rho} \theta^\rho (-2\delta_{\beta\mu}^{\alpha\tilde{\nu}}) \lambda_{\tilde{\nu}} \hat{\lambda}^{\tilde{B}} A_{\alpha \tilde{B}} \\ & = (-\frac{1}{4}) \lambda^B (\gamma_m)_B \gamma^\theta \partial_\theta (\gamma^m)_{\delta\rho} \theta^\rho (\delta_B^\alpha \delta_m^{\tilde{\nu}} - \delta_B^{\tilde{\nu}} \delta_m^\alpha) \lambda_{\tilde{\nu}} \hat{\lambda}^{\tilde{B}} A_{\alpha \tilde{B}} \\ & = (-\frac{1}{4}) \left[\lambda^\alpha (\gamma_m)_B \gamma^\theta \partial_\theta (\gamma^m)_{\delta\rho} \theta^\rho \lambda_{\tilde{\nu}} \hat{\lambda}^{\tilde{B}} A_{\alpha \tilde{B}} \right. \\ & \quad \left. - \lambda^{\tilde{\nu}} (\gamma_m)_B \gamma^\theta \partial_\theta (\gamma^m)_{\delta\rho} \theta^\rho \lambda_{\tilde{\nu}} \hat{\lambda}^{\tilde{B}} A_{\alpha \tilde{B}} \right] \\ & = \frac{1}{4} \left[\lambda^\alpha (\gamma_m)_B \gamma^\theta \partial_\theta (\gamma^m)_{\delta\rho} \theta^\rho \lambda_{\tilde{\nu}} \hat{\lambda}^{\tilde{B}} A_{\alpha \tilde{B}} \right. \\ & \quad \left. + \lambda^2 (\gamma_m)_B \gamma^\theta \partial_\theta (\gamma^m)_{\delta\rho} \theta^\rho \hat{\lambda}^{\tilde{B}} A_{\alpha \tilde{B}} \right] \end{aligned}$$

$$\sim \Delta A^{(p)} \rightarrow F^{(p+1)} = dA^{(p)} \quad D=10$$

$$* F^{(p+1)} = G^{10-p-1} = dB^{10-p-2} \quad p=3$$

Where the hell does he get a 3 form from?

For the above scattering we integrate (Ref 43) over the zero modes:

$$\begin{aligned} a & \propto \langle V_1 V_2 V_3 \int \mathcal{V}_1 \dots \mathcal{V}_n \rangle \\ & = \int d^{10}x \int d^{10}\lambda \int d^4\theta \lambda^\alpha \lambda^{\tilde{\beta}} \lambda^\gamma f_{\alpha\beta\gamma}(x, \theta, \lambda) \end{aligned}$$

~> Comes from the fact that $Q(\lambda^\alpha \lambda^{\tilde{\beta}} \lambda^\gamma f_{\alpha\beta\gamma}) = 0$ when we are on-shell.

$\langle Y \rangle = 0$ trivially from $Y = Q\Omega$ i.e it is physical ($Q|0\rangle = 0$)

• For the Twistor particle we had, when looking at Chern-Simons in 3D the Superfield was $\hat{\mu} \hat{\Gamma}$ with highest ghost weight

$$\sim \frac{i}{6} \epsilon_{\mu\nu\rho} C^\mu C^\nu C^\rho C(x)$$

$$\text{with } \langle C^\mu C^\nu C^\rho \rangle = i \epsilon^{\mu\nu\rho}$$

• For the " " in 10D the highest ghost states had

$$\langle (\lambda \gamma^m \theta) (\lambda \gamma^n \theta) (\lambda \gamma^p \theta) (\theta \gamma_{mnp} \theta) \rangle = 1$$

to normalize our Chern Simons Action

~> This cohomology class contained only one state

and this also holds true for the string

$$f_{\alpha\beta\gamma} = A_{\alpha\beta\gamma} + \theta^\delta B_{\alpha\beta\gamma\delta} + \dots + (\gamma^m \theta)_\alpha (\gamma^n \theta)_\beta (\gamma^p \theta)_\gamma (\theta \gamma_{mnp}) F$$

So the "top form" isomorphism yields

$$\left[\begin{aligned} & \left(\frac{\partial}{\partial \theta} \gamma_m \right)^\alpha = \frac{\partial}{\partial \theta} (\gamma_m)^\alpha \rightarrow (D \text{ spin } 1 (\gamma^m \theta)_\beta \dots F) \\ & \Rightarrow \frac{\partial}{\partial \theta} (\gamma_m)^\alpha \delta (f_{\alpha\beta\gamma}) \sim (\gamma_m)^{\alpha\delta} (\gamma_{\alpha\delta} (\gamma^m \theta)_\beta) F \dots \\ & \text{(i)} + (\gamma_m)^{\alpha\delta} (\gamma^m \theta)_\alpha (\gamma^n)_{\beta\delta} [\dots] F \\ & \text{(ii)} + (\gamma_m)^{\alpha\delta} (\gamma^m \theta)_\alpha (\gamma^n \theta)_\beta (\gamma^p)_{\gamma\delta} (\theta \dots) F \\ & \text{(iii)} + (\gamma_m)^{\alpha\delta} (\dots) (\gamma_{mnp} \theta)_\delta F \\ & \text{(i)} (\gamma_m)^{\alpha\delta} (\gamma^m \theta)_\alpha (\gamma^n)_{\beta\delta} \dots = (\gamma_m)^{\alpha\delta} (\gamma^m_{\alpha\delta} \theta^2) (\gamma^n)_{\beta\delta} \\ & = D \delta^\delta_\epsilon \theta^2 (\gamma^n)_{\beta\delta} [\dots] F \\ & \text{(ii)} D \theta^\delta \dots \\ & \text{(iii)} D \theta^\delta (\dots) (\gamma_{mnp} \theta)_\delta \end{aligned} \right.$$

$\Rightarrow \int d\theta^{\alpha\beta\gamma}$ must have a normalization of $\sim \frac{1}{D}$ (Paper does not make it explicit)
(here $D=10$) (Also Do not make $\delta^D(\Sigma)$ explicit)

$$\text{Full form: } \langle V_1 V_2 V_3 | u_1 \dots u_n \rangle = \delta^{10}(\Sigma, k_i) \left(\frac{\delta}{\delta \theta} \right)^{\alpha\beta\gamma} f_{\alpha\beta\gamma}$$

$$\text{where } \left(\frac{\delta}{\delta \theta} \right)^{\alpha\beta\gamma} \equiv \left(\frac{\partial}{\partial \theta} \gamma_m \right)^\alpha \left(\frac{\partial}{\partial \theta} \gamma_n \right)^\beta \left(\frac{\partial}{\partial \theta} \gamma_p \right)^\gamma \left(\frac{\partial}{\partial \theta} \gamma_{mnp} \theta \right) \Big|_{\theta=0}$$

* For Bosonic String theory we have $\langle C \partial C \partial C \rangle$ (ghost order 3)

* 3pt amplitude ("free")

$$a_3 = \langle V_1 V_2 V_3 \rangle = \langle \lambda^\alpha A_{1\alpha} \lambda^\beta A_{2\beta} \lambda^\gamma A_{3\gamma} \rangle$$

$$A_\alpha = \frac{1}{2} (\gamma^m \theta)_\alpha a_m + \frac{1}{12} (\theta \gamma_{mnp} \theta) (\gamma_{mnp})_{\alpha\beta} \gamma^\beta + \dots$$

$a_m = G_{\mu\nu}$

~ In $D=10$ we have the decomposition

$$f_{\alpha\beta} = \gamma_{\alpha\beta} f_m + \gamma_{\alpha\beta}^{\mu\nu\rho\sigma} f_{\mu\nu\rho\sigma}$$

~ For Quark Scattering in $N=4$ YM we usually have

$$A_n^{\text{tree}}[123] = \delta^{(2)}(\tilde{Q}) / \langle 12 \rangle \langle 23 \rangle \dots \langle n1 \rangle$$

since $D=10, N=16$

↓

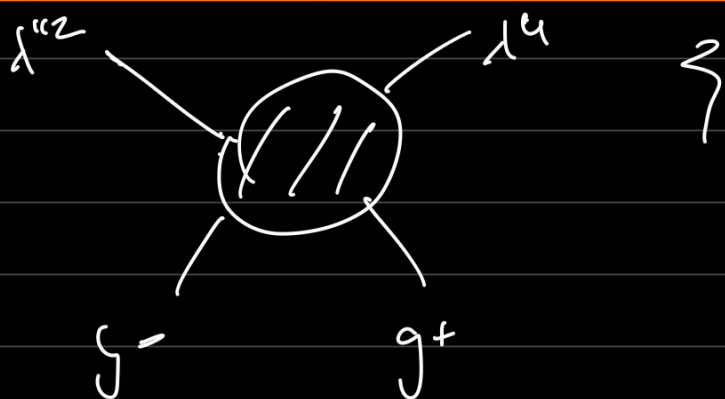
$D=4, N=4$ SPM

$$\delta^{(3)}(\vec{Q}) = \frac{1}{24} \frac{1}{11} \sum_{A=1}^4 \sum_{i,j=1}^4 \langle ij \rangle \mathcal{N}_A^{(i)} \mathcal{N}_A^{(j)}$$

• Look at n -point Scattering:

g^- : gluon ($h < 0$) λ^{123} : Gluino ($h < 0$)
 g^+ : gluon ($h > 0$) λ^4 : Gluino ($h > 0$)

$$A_n[g^-, \lambda^{123}, \lambda^4, g^+, \dots, g^+] = \frac{\langle 13 \rangle}{\langle 12 \rangle} A[g^-, g^-, g^+, \dots, g^+]$$



$$\hookrightarrow A_n[g^-, \lambda^{123}, \lambda^4, g^+, \dots, g^+] = (\partial_1)^4 (\partial_1^{(2)} \partial_2^{(2)} \partial_3^{(2)}) (\partial_4^{(3)}) A_n[\dots]$$

$$= (\partial_1)^4 (\partial_1^{(2)} \partial_2^{(2)} \partial_3^{(2)}) (\partial_4^{(3)}) \left(\frac{1}{24} \frac{1}{11} \sum_{A=1}^4 \sum_{i,j=1}^4 \langle ij \rangle \mathcal{N}_A^{(i)} \mathcal{N}_A^{(j)} \right)$$

$$= \frac{1}{24} (\partial_1)^4 (\partial_1^{(2)} \partial_2^{(2)} \partial_3^{(2)}) \left(\sum_{i,j} \langle ij \rangle \right) (\mathcal{N}_1^{(1)} \mathcal{N}_1^{(5)} \mathcal{N}_2^{(1)} \mathcal{N}_2^{(1)} \dots \mathcal{N}_4^{(1)} \mathcal{N}_4^{(5)})$$

$$\rightarrow \partial_4^{(3)} = \mathcal{N}_1^{(1)} \mathcal{N}_1^{(5)} \dots (\delta^{3i} \mathcal{N}_4^{(i)} - \mathcal{N}_4^{(i)} \delta^{3i})$$

$$(\partial_1)^4 (\partial_1^{(2)} \partial_2^{(2)}) \left(\sum_{i,j} \langle ij \rangle \mathcal{N}_1^{(i)} \mathcal{N}_1^{(j)} \dots (\delta^{2i} \mathcal{N}_1^{(i)} - \mathcal{N}_1^{(i)} \delta^{2i}) \right)$$

$$= (\partial_1)^4 (\partial_1^{(2)} \sum_{i,j} \langle ij \rangle \mathcal{N}_1^{(i)} \mathcal{N}_1^{(j)} (\delta^{2i} \mathcal{N}_2^{(i)} - \mathcal{N}_2^{(i)} \delta^{2j}) (\dots))$$

$$\frac{1}{24} (\partial_1)^4 \sum_{i,j} \langle ij \rangle (\delta^{2i} \mathcal{N}_1^{(j)} - \mathcal{N}_1^{(i)} \delta^{2j}) (\dots) (\delta^{3i} \mathcal{N}_4^{(j)} - \mathcal{N}_4^{(i)} \delta^{3j})$$

$$= \frac{1}{24} \sum_{i,j} \langle ij \rangle (\delta^{2i} \delta^{j1} - \delta^{i1} \delta^{2j}) (\delta^{2i} \delta^{j1} - \delta^{i1} \delta^{2j})$$

$$= \frac{1}{24} \sum (\delta^{2i} \delta^{j1} - \delta^{i1} \delta^{2j})^3 \langle ij \rangle^3 (\delta^{3i} \delta^{j1} - \delta^{i1} \delta^{3j}) \langle ij \rangle$$

$$\text{each yield} \sim \frac{2^4}{24} \sum_{i,j} (\delta^{2i} \delta^{j1}) \langle i j \rangle^3 (\delta^{3i} \delta^{j1}) \langle i j \rangle$$

$$= \langle 12 \rangle^3 \langle 13 \rangle$$

$$\begin{aligned} \vec{o} \circ A_n [g^- \chi^{123} \chi^4 g^+ \dots g^+] &= \frac{\langle 12 \rangle^3 \langle 13 \rangle}{\langle 12 \rangle \langle 23 \rangle \dots \langle n1 \rangle} \\ &= \frac{\langle 13 \rangle}{\langle 12 \rangle} \left(\frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \dots \langle n1 \rangle} \right) \\ &= \frac{\langle 13 \rangle}{\langle 12 \rangle} A_n [g^- g^- g^+ \dots g^+] \quad \square \end{aligned}$$

* Look at the 3pt: Take 3-Gluon
anti-MHV: $A_3 [g^- g^+ g^+]$

$$A_3^{\text{anti-MHV}} = \frac{\delta^{(4)}([12]n_3 + [23]n_1 + [31]n_2)}{[12][23][31]} \quad \tau \leftarrow +$$

$$\delta^{(4)}(\sim) = \prod_{A=1}^4 ([12]n_A^{(1)} + [23]n_A^{(1)} + [31]n_A^{(2)})$$

$$\begin{aligned} \Rightarrow A_3 [g^- g^+ g^+] &= (\partial_3)^4 A_3^{\text{anti-MHV}} \\ &= (\partial_1^1 \partial_2^1 \partial_3^1) = \prod_{A=1}^4 ([12] \delta^{13} + [23] \delta^{11} + [31] \delta^{21}) \\ &= \prod_{A=1}^4 ([12] \delta^{3A} + [23] \delta^{1A} + [31] \delta^{2A}) \\ &= \frac{[23]^4}{[12][23][31]} = \frac{[23]^3}{[12][31]} = 1 \dots \end{aligned}$$

3-pt Gluon Scattering

$$\begin{aligned} \text{* For MHV: } A_3 [g^+ g^- g^-] &= A^* [g^- g^+ g^+] \\ &= \frac{\langle 23 \rangle^3}{\langle 12 \rangle \langle 31 \rangle} \end{aligned}$$

* Scalar 4 pt Scattering
in $N=4$

$$A_4 [s^{12} s^{34} s^{12} s^{34}] = (\partial_1^1 \partial_2^1) (\partial_3^2 \partial_4^2) (\partial_1^3 \partial_2^3) (\partial_3^4 \partial_4^4) A$$

$$= \frac{\langle 13 \rangle \langle 13 \rangle \langle 24 \rangle \langle 24 \rangle}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle}$$

• That's fine: This is a Scalar QED theory

$$A_{\text{tree}}^{\text{BCFW}}(\varphi\varphi^*\varphi\varphi^*) = e^2 \left(\frac{\langle 13 \rangle^2 \langle 24 \rangle^2}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle} \right)$$

↳ Note: The $N=4$ Scalar scattering has given us the BCFW Amplitude and not the Feynman Amplitude

$$A_{\text{tree}}^{\text{Fey}}(\varphi\varphi^*\varphi\varphi^*) = e^2 \left(1 + \frac{\langle 13 \rangle^2 \langle 24 \rangle^2}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle} \right)$$

Scalar QED:

$$\mathcal{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} \sim |\partial\varphi|^2 + ieA^\mu [\partial_\mu\varphi^*]\varphi - \varphi^* \partial_\mu\varphi] - e^2 A^\mu A_\mu \varphi\varphi$$

$$\sim \frac{1}{4} \lambda |\varphi|^4$$

$$A_4(\varphi\varphi^*\varphi\varphi^*) : \sim (ie)^2 \langle \varphi_1 \varphi_2^* \varphi_3 \varphi_4^* \int d^4\omega d^4z A^\mu \dots \rangle$$

$$- \frac{1}{4} i \lambda \langle \varphi_1 \varphi_2^* \varphi_3 \varphi_4^* \int d^4\omega |\varphi|^4 \rangle$$

e^2 term: $\sim 2ie^2 \mathcal{M}_{\mu\nu}$ vertex

$$\partial_\mu\varphi = \int d^4p a^\dagger i p_\mu e^{ipx} + \dots$$

$\Rightarrow$ Vertex terms with $\varphi\partial\varphi \sim p_\mu$

$$: (ie)^2 \int d^4\omega d^4z \Delta(x_1-z) \Delta(x_2-z) \Delta(x_3-\omega) \Delta(x_4-\omega)$$

$$\sim \frac{1}{i!} \int \frac{d^4p_i}{(2\pi)^3} \int d^4\omega d^4z \left(\frac{1}{p_i^2 + m^2 - i\epsilon} \right) \delta \dots$$

$$\Rightarrow E_z^\mu E_{\mu-\omega}(p,q) = \frac{2\langle 13 \rangle [24]}{2[12][34]} = \frac{\langle 13 \rangle [24]}{[12][34]}$$

$$\langle 12 \rangle [12] = (p_1 + p_2)^2$$

by conservation of momenta

$$\sum_i |i\rangle [i] = 0$$

$$\hookrightarrow \sim \left(\frac{\langle 13 \rangle [24]}{[12][34]} \right) \left(\frac{1}{\langle 12 \rangle [12]} \right) \left(\frac{1}{\langle 34 \rangle [34]} \right)$$

$$\langle 31 \rangle [1] + \langle 32 \rangle [2] + \langle 34 \rangle [4] = 0$$

$$\hookrightarrow \frac{\langle 13 \rangle^2 [24]^2}{[12][34]}$$

$$\rightarrow A_4(\varphi\varphi^*\varphi\varphi^*) = -\lambda + \hat{e}^2 \left(1 + \frac{1}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle} \right)$$

* BCFW computes the case in which
 $\lambda = \hat{e}^2$

- For general scattering: for λ in 3pt there is no way the theory can know the contents of $\lambda[k]^4$
 - $\rightarrow$ This is gauge-invariant info.
 - $\rightarrow$ For $N=4$: this symmetry is fixed
- * Can define Super-BCFW (at tree level) recursive relations.

*